

AINSWORTH, NE, USA

WMO: 725556

Lat: 42.577N

Lon: 100.001W

Elev: 2584

StdP: 13.37

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.0	(c) -0.2	(d) -13.0	(e) 3.0	(f) -5.1	(g) -7.7	(h) 4.0	(i) 0.4	(j) 34.4	(k) 22.7	(l) 31.6	(m) 24.3	(n) 9.2	(o) 320	(p) 0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	95.1	69.0	91.0	69.0	88.0	68.5	74.3	85.5	72.6	83.9	71.1	82.1	13.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	129.0	78.5	69.8	120.6	77.8	67.8	112.6	76.8	39.7	85.5	38.1	83.8	36.6	82.5	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 29.9	(b) 26.2	(c) 23.5	(e) -14.3	(f) 100.0	(g) 4.6	(h) 4.7	(i) -17.6	(j) 103.4	(k) -20.3	(l) 106.1	(m) -22.8	(n) 108.8	(o) -26.2	(p) 112.2		
(a) 29.9	(b) 26.2	(c) 23.5	(e) -14.2	(f) 76.9	(g) 4.1	(h) 1.6	(i) -17.1	(j) 78.0	(k) -19.6	(l) 78.9	(m) -21.9	(n) 79.8	(o) -24.9	(p) 80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	25.3	27.5	37.2	46.7	57.1	67.4	74.7	71.8	63.6	49.5	37.4	26.8
	DBStd	20.01	12.84	12.72	12.35	10.07	8.75	7.03	6.21	5.91	8.69	9.87	11.29	11.72
	HDD50	3276	764	632	422	174	32	1	0	0	6	128	397	720
	HDD65	6666	1229	1050	862	551	273	55	6	13	128	486	828	1185
	CDD50	2864	0	2	25	76	252	522	765	676	414	113	19	0
	CDD65	776	0	0	0	2	27	125	306	224	86	6	0	0
	CDH74	8434	0	0	21	85	393	1314	3325	2127	1018	138	13	0
	CDH80	3473	0	0	2	17	114	448	1572	884	399	36	1	0
(14) Wind	WSAvg	10.8	11.0	11.3	11.7	12.5	11.7	10.2	9.3	9.2	10.3	10.8	10.9	10.8
(15) Precipitation	PrecAvg	22.0	0.4	0.6	1.1	2.7	3.6	3.9	2.9	2.1	1.9	1.6	0.7	0.5
	PrecMax	28.2	1.3	2.2	2.6	6.4	7.1	8.1	5.5	3.8	5.4	3.6	2.1	1.5
	PrecMin	11.6	0.0	0.1	0.1	0.6	0.5	0.7	0.6	0.2	0.4	0.0	0.0	0.0
	PrecStd	4.2	0.3	0.4	0.6	1.3	1.9	1.7	1.5	0.9	1.1	1.1	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.8	65.7	77.0	82.1	88.5	94.6	102.2	98.6	94.6	84.4	75.0	60.9
		MCWB	43.8	47.0	52.4	56.6	64.9	69.5	68.8	69.1	64.8	59.7	51.8	45.4
	2%	DB	52.4	56.7	69.8	75.2	82.6	89.1	97.1	92.7	89.8	77.4	66.0	53.1
		MCWB	41.0	43.4	50.9	54.5	63.1	69.4	70.3	69.1	66.3	56.9	49.0	41.4
	5%	DB	46.2	50.1	62.9	70.1	78.6	85.0	92.6	89.2	84.3	72.1	59.5	46.2
		MCWB	37.7	40.0	48.4	52.9	61.4	67.7	70.3	69.1	64.9	55.4	45.4	37.7
	10%	DB	41.9	44.7	55.5	64.1	73.3	81.7	89.3	84.4	80.6	66.0	54.0	41.4
		MCWB	35.4	37.4	44.9	50.5	58.7	66.2	70.3	68.5	63.2	52.8	43.4	34.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	48.7	57.4	60.0	68.5	74.8	77.1	75.4	72.1	63.2	53.7	46.5
		MCDB	58.1	62.2	65.2	74.4	81.6	85.2	88.0	85.9	83.1	75.0	68.2	58.4
	2%	WB	41.4	44.1	52.6	56.8	65.8	72.2	74.9	73.2	69.3	60.2	50.0	41.8
		MCDB	51.7	55.3	65.9	71.1	78.5	83.3	86.3	83.8	81.3	71.8	63.4	51.8
	5%	WB	38.4	40.5	49.1	54.2	63.2	70.3	73.3	71.6	67.5	57.1	46.7	38.4
		MCDB	45.7	49.6	62.4	67.1	74.8	81.2	85.1	81.9	79.3	68.4	59.5	46.2
	10%	WB	35.6	37.4	45.3	51.6	60.6	68.2	71.7	70.2	65.5	54.0	43.4	35.3
		MCDB	41.1	44.0	56.0	62.7	70.7	79.0	83.4	80.2	76.4	64.5	52.6	40.7
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	20.3	22.0	22.2	21.5	22.0	23.3	22.7	24.5	23.2	21.9	19.1
		MCDBR	26.2	28.4	33.6	33.0	29.2	27.1	29.2	29.0	30.7	32.8	31.2	26.0
		MCWBR	16.8	16.8	17.5	16.3	13.5	11.7	9.8	10.6	11.6	15.7	16.6	16.7
		MCDBR	25.0	27.0	30.7	29.0	25.1	23.8	24.3	23.7	24.9	28.1	30.1	25.2
	5% WB	MCWBR	16.5	16.5	16.9	15.2	12.7	12.0	11.1	10.6	11.6	15.5	16.5	16.6
(40) Clear-Sky Solar Irradiance	taub		0.260	0.270	0.294	0.330	0.350	0.365	0.383	0.385	0.344	0.312	0.271	0.260
	taud		2.528	2.510	2.484	2.386	2.417	2.406	2.393	2.366	2.445	2.507	2.544	2.513
	Ebn at Noon		287	301	304	298	293	287	281	277	282	280	280	276
	Edh at Noon		22	26	30	36	36	37	37	37	31	26	21	20
(44) All-Sky Solar Radiation	RadAvg		661	946	1276	1606	1901	2166	2202	1871	1517	1046	724	558
	RadStd		50	63	107	92	153	139	76	109	94	109	53	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page